



ROOT CAUSE UNLOCKED

Unlock Your *ADHD*

A Functional Medicine Roadmap to Real, Lasting Focus

For people who are stuck and ready for a breakthrough.

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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BEFORE YOU START

Contents

A Note Before You Begin

Before Anything Else: What Is Not a Supplement Question

On Stimulant Medication: Read This First

Why This Guide Exists

KEY 1

The Root Cause Conversation

KEY 2

The ADHD Diet

KEY 3

Lab Testing That Actually Answers Questions

KEY 4

Nutritional Supplementation

KEY 5

Therapeutic Modalities

KEY 6

Working With a Practitioner

KEY 7

Putting It All Together

Final Thought

Your ADHD Tracker

Working With Root Health

APPENDIX

Quick Reference

References

A Note Before You Begin

This book is written for educational purposes and is not intended to replace the advice of a qualified healthcare professional. It does not diagnose ADHD or any other condition. The information here reflects a functional medicine perspective focused on root cause identification. Please work with a licensed provider before changing, stopping, or starting any supplement or treatment protocol.

That said, the information in these pages is evidence-informed and immediately actionable. It is the same framework I use with patients at Root Health every single day.

Before Anything Else: What Is Not a Supplement Question

Almost everything in this book is meant for symptoms that have built up gradually over months or years. A few situations are different and belong with a physician the same day, not with a protocol.

If focus, memory, or personality changed suddenly over hours or days, if the change followed a head injury or a fall, if it came on alongside fever or confusion, or if it started right after a new prescription or a dose change, that is a call to your physician or a trip to urgent care, not a nutrient plan.

If you are having thoughts of harming yourself or of suicide, please stop reading and reach out right now. Call or text **988** to reach the Suicide and Crisis Lifeline in the United States, available 24 hours a day, every day. If you or someone else is in immediate danger, call **911** or go to your nearest emergency department. You do not need a diagnosis or a good enough reason to reach out. Everything else in this book can wait. That cannot.

On Stimulant Medication: Read This First

If you take a stimulant or any other prescription for ADHD, nothing in this book is a reason to stop it, skip a dose, or change how you take it. Starting, adjusting, and stopping a prescription belongs to you and the physician managing that part of your care. Prescribing is not something I do, and this book is written to run alongside that care, not instead of it.

That boundary is not a formality and it is not me being careful for the sake of it. Everything in these pages works fine next to a prescription. If a change to yours ever makes sense, the person who wrote it is the person to make that decision with.

Why This Guide Exists

If you are reading this, you have probably tried things. Maybe you have been on medication for years and wonder if there is something underneath the diagnosis that nobody has looked at. You might not want medication at all and be searching for a path that makes sense. Or you have been told your brain just works differently and that you should learn to live with it.

This guide takes a different position. ADHD is real. But the symptoms you are living with, the inability to focus, the restlessness, the brain fog, the emotional dysregulation, those things do not always exist in isolation from what is happening in the rest of your body. Your brain does not float separate from your gut, your liver, your immune system, or your hormones. It is connected to all of it, every hour of every day.

What functional medicine does is look at that whole picture. It asks why a nervous system becomes dysregulated in the first place. It asks what the body needs that it is not getting, and what it is being burdened with that it cannot process. This guide will walk you through the key areas that matter, give you specific tools you can start using right now, and show you what a truly comprehensive approach to ADHD looks like.

You do not have to accept that this is just how you are forever. You have to be willing to look deeper than the symptoms.

What the Evidence Behind This Guide Actually Is, Stated Plainly

I would rather you know where you are standing than be sold a uniform level of certainty across seven chapters, because that certainty does not exist.

The strongest evidence in this book sits behind the least exciting parts. Nutrition, exercise, sleep, and neurofeedback have been studied in ADHD in randomized trials and systematic reviews, and I cite that work where I lean on it. The functional testing and the targeted supplement protocol sit at a different tier: laboratory and mechanistic evidence, plus reasoned clinical practice, rather than human outcome trials for this exact protocol in this exact population. That is a real distinction and I will name it as we go rather than blurring it.

What keeps the second tier honest is the endpoint. I do not ask you to believe a protocol works because the mechanism is elegant. I ask you to log a baseline before you change anything, and then compare yourself to your own log. If the numbers you track do not move, the protocol is wrong for you and we change it. That is the standard I hold my own work to, and it is the standard this book asks you to hold it to.

The Root Cause Conversation

What Conventional Medicine Gets Right and Wrong

Stimulant medications work for many people.^{1,2} This guide is not here to argue against that. What they do is increase the availability of dopamine and norepinephrine in the brain, which temporarily improves the signal-to-noise ratio that makes focus hard. For some, that is genuinely life changing.

Where conventional care often falls short is in stopping there. When a medication stops working, the dose goes up. When side effects become a problem, a different medication gets tried. The question of why the brain is running low on these neurotransmitters in the first place rarely gets asked.

Functional medicine asks that question. Dopamine is made from tyrosine. Norepinephrine is made from dopamine. These are nutritional processes. They depend on specific nutrients, enzyme function, gut health, and freedom from toxins that block those pathways. When those upstream factors are not addressed, you are always managing symptoms rather than restoring function.

The Systems That Drive ADHD Symptoms

ADHD symptoms rarely come from a single broken pathway. The people who struggle the most tend to have several systems out of balance at the same time. The most common ones are:

- Neurotransmitter production and metabolism, particularly dopamine, norepinephrine, and serotonin pathways
- Gut health and the gut-brain axis, which directly influences inflammation, mood, and cognitive clarity
- Methylation, the biochemical process involved in neurotransmitter synthesis and gene regulation
- Nutrient status, especially magnesium, zinc, iron, omega-3 fatty acids, and B vitamins
- Toxic burden from heavy metals, environmental chemicals, and mycotoxins that impair brain function
- Blood sugar regulation, because the brain is exquisitely sensitive to glucose instability
- Mitochondrial function, since every neuron depends on adequate energy production to fire correctly

How I Approach It

I do not guess. I test. I look at what is actually happening in your biochemistry and use that information to build a protocol that addresses the real drivers of your symptoms. The chapters that follow will show you exactly what that looks like and how you can begin applying these principles right now.

The ADHD Diet

Food Is Information

Before any supplement, before any test, before any therapy device, there is food. What you eat every day either supports your nervous system or burdens it. Food is the foundation that everything else gets built on.

The ADHD brain is hungry. It burns through nutrients faster than average. It is more sensitive to blood sugar crashes. It is more reactive to food-based inflammatory triggers. Getting the diet right does not resolve ADHD on its own. Getting it wrong, though, makes everything else much harder to move.

What to Remove First

The biggest dietary wins for ADHD typically come from removing things rather than adding them. The following are the highest-priority targets.

Refined Sugar and High-Glycemic Carbohydrates

Blood sugar swings are brutal for the ADHD brain. When glucose spikes and then crashes, the brain goes into a kind of stress response. Cortisol rises, adrenaline kicks in, and focus and emotional regulation go out the window. That cycle also puts constant demand on the stress-response system, which worsens the underlying dysregulation over time.

Removing juice, soda, candy, white bread, and processed snacks and replacing them with low-glycemic whole foods is one of the highest-impact steps a person with ADHD can take. This alone improves cognitive function for many people within two to three weeks.

Artificial Dyes and Additives

The research connecting artificial food dyes, particularly Red 40, Yellow 5, and Yellow 6, to increased hyperactivity and impulsivity is among the more consistently replicated findings in the diet and behavior literature. The UK required warning labels on products containing these dyes years ago. The United States has been slower to act, but the evidence is strong enough that these ingredients deserve to be removed from the diet entirely. Check labels on anything processed.

Gluten and Dairy

A significant subset of people with ADHD react to gluten or dairy, and many others are unaffected. The gut-brain connection means that intestinal inflammation triggered by these foods can affect neurotransmitter production and cognitive function. A 60-day trial eliminating both, done properly, is often more informative than any food sensitivity test. Many people report clearer thinking within the first two weeks.

Pesticide Exposure

Organophosphate pesticides found on conventional produce have been linked to increased rates of ADHD in children and to ongoing cognitive disruption in adults. Switching to organic produce, especially for the highest-pesticide items on the Environmental Working Group dirty dozen list, is a meaningful change with low cost and high return.

What to Build the Diet Around

Protein at Every Meal

Protein provides the amino acids that become neurotransmitters. Tyrosine becomes dopamine. Tryptophan becomes serotonin. Without adequate protein throughout the day, you are simply running low on the raw materials your brain needs to function. Eggs, pasture-raised meat, wild-caught fish, legumes, and nuts should anchor every meal.

Omega-3 Fatty Acids

Lower omega-3 status shows up more often in people with ADHD than in the general population. DHA in particular is a primary structural component of brain cell membranes and is involved in the function of dopamine receptors. Cold-water fatty fish like wild salmon, sardines, and mackerel are the best food sources. Walnuts, flaxseed, and chia seeds provide ALA, though conversion to DHA is limited and supplementation is typically needed on top of dietary sources.

Here is the honest evidence picture, because omega-3 is one of the most oversold supplements in this category. Randomized trial evidence for fish oil as a stand-alone treatment for core ADHD symptoms is mixed rather than settled, and the major independent reviews of complementary approaches say so plainly.^{3,4} It is still worth taking, as long as you know what it is for. I use omega-3 as structural and anti-inflammatory support for a brain that is being asked to rebuild, not as a substitute for a stimulant, and I judge it by whether your own tracked focus, mood, and sleep move, not by the promise on the bottle.

Magnesium-Rich Foods

Magnesium is involved in hundreds of enzymatic reactions in the body and is one of the most common nutrient deficiencies I see in people with ADHD. It is required for the production of ATP, the regulation of the nervous system, and the synthesis of neurotransmitters. Dark leafy greens, pumpkin seeds, black beans, avocado, and dark chocolate are among the best dietary sources.

Fermented Foods and Gut Support

The gut produces the large majority of the serotonin in the body. A compromised microbiome compromises neurotransmitter production and drives the neuroinflammation that makes ADHD symptoms worse. Including fermented foods like sauerkraut, kimchi, kefir, and plain yogurt with live cultures on a daily basis supports the microbial diversity that keeps this system working correctly.

Zinc-Rich Foods

Zinc plays a direct role in dopamine metabolism and is frequently low in people with ADHD. Oysters are by far the highest food source, followed by beef, pumpkin seeds, chickpeas, and cashews. Zinc has been studied in ADHD both as an adjunct to stimulant medication and on its own, with some trials reporting meaningful improvement in some populations. The evidence base is modest rather than decisive, which is exactly why I test zinc status rather than assume it.

Quick Start: The ADHD Plate

- **Half the plate.** Non-starchy vegetables and leafy greens
- **One quarter.** Quality protein (eggs, fish, poultry, meat, legumes)
- **One quarter.** Complex carbohydrates (sweet potato, brown rice, quinoa, squash)
- **Add.** Healthy fat at every meal (avocado, olive oil, nuts, seeds)
- **Drink.** Water and herbal teas. Minimize caffeine after noon. Avoid all soda and juice.

Lab Testing That Actually Answers Questions

The “Don’t Guess, Test” Philosophy

One of the most frustrating things about managing ADHD through conventional medicine is that most treatment decisions are made without any biochemical data. A questionnaire gets filled out. A diagnosis gets made. A prescription gets written. Nobody looks inside.

Functional medicine takes the opposite approach. Before making specific nutritional or supplemental recommendations, you need to know what is actually happening in the body. That means knowing what your gut microbiome is doing, which neurotransmitter pathways look disrupted, whether heavy metals or toxins are affecting brain function, and which nutrients you are actually depleted in.

One framing note before the table, because it matters. These are functional panels. They are suggestive rather than diagnostic, and they are most useful for showing where to look and what to address first, held next to your history and your exam. A result on any of them is consistent with a pattern. It does not confirm a diagnosis, and I do not use it that way.

The tests below are the ones I find most clinically useful for people dealing with ADHD. They are available through Vibrant Wellness, Genova Diagnostics, and Diagnostic Solutions Laboratory, and can be ordered through a functional medicine provider.

The Core Testing Panel

Lab test	Lab company	What it reveals	Why it matters for ADHD
Organic Acids Test (OAT), the primary Vibrant option	Vibrant Wellness / Genova	Neurotransmitter metabolites, mitochondrial function, gut dysbiosis markers, nutrient status, and toxic metabolites. The Vibrant OAT is the preferred option when already using Vibrant panels. If using Genova, the NutrEval below covers this ground and more.	Provides a detailed window into which brain chemistry pathways are under stress, and offers meaningful clinical direction for why focus and regulation are struggling
Gut Zoomer, choose one: best for microbiome terrain	Vibrant Wellness	Over 300 microbiome species, gut permeability markers (zonulin), secretory IgA, inflammation, parasites, yeast overgrowth. Choose this when the primary question is microbiome terrain and gut-brain axis disruption.	Gut health is brain health. Dysbiosis and increased permeability drive neuroinflammation that worsens every ADHD symptom. The strongest option when overall microbial ecology and permeability are the primary concern.
GI-MAP, choose one: best for pathogen precision	Diagnostic Solutions	Comprehensive PCR-based stool test covering pathogens (with H. pylori virulence factor detail), bacteria, yeast, parasites, and digestive function markers including elastase and calprotectin. Choose this when there is strong pathogen suspicion or active gastrointestinal symptoms alongside ADHD.	When the clinical picture points toward a specific pathogen, significant digestive dysfunction, or gastrointestinal symptoms driving the neurological presentation, GI-MAP gives more granular data than Gut Zoomer. Run one or the other, not both.
Micronutrient Panel	Vibrant Wellness / Genova	Functional intracellular levels of more than 40 vitamins, minerals, amino acids, and antioxidants	Serum levels miss deficiencies inside the cell. This test shows what your brain actually has access to
Neural Zoomer Plus	Vibrant Wellness	Antibodies to neurological tissues including the blood-brain barrier, myelin, and key neurological proteins	Identifies immune reactivity to brain tissue, which can drive cognitive symptoms indistinguishable from ADHD
Total Tox Burden	Vibrant Wellness	Heavy metals, mycotoxins, and environmental chemicals in one panel	Toxic burden is a major and underappreciated driver of ADHD

Lab test	Lab company	What it reveals	Why it matters for ADHD
			symptoms. Lead and mercury impair dopamine function.
NutrEval, the Genova comprehensive alternative to OAT	Genova Diagnostics	Covers the same organic acids territory as the OAT, plus plasma amino acid profiles, fatty acid panels, and direct serum nutrient markers. Amino acid levels matter particularly for ADHD, since tyrosine, tryptophan, and glutamine are direct neurotransmitter precursors that OAT metabolites alone cannot fully characterize.	If using Genova, choose NutrEval instead of a stand-alone OAT. It covers everything the OAT covers and more. If already running Vibrant panels, use the Vibrant OAT and skip NutrEval to avoid overlapping data.

What These Tests Reveal for ADHD

Organic Acids and Neurotransmitter Metabolism

The Organic Acids Test is one of the most useful tests available for understanding ADHD from a biochemical standpoint. It measures metabolites that reflect the function of neurotransmitter pathways rather than just the presence of certain compounds in the blood. If you are building a panel primarily through Vibrant Wellness, the stand-alone OAT is the right choice. If you are working through Genova Diagnostics, the NutrEval covers everything the OAT covers and adds plasma amino acid and fatty acid data that are clinically meaningful for ADHD, making it the more comprehensive single test on that platform. Running both an OAT and a NutrEval at the same time produces heavily overlapping data and is not necessary.

Specifically, it looks at HVA (homovanillic acid) and VMA (vanillylmandelic acid), which are dopamine and norepinephrine breakdown products. Low levels of these metabolites are consistent with inadequate dopamine and norepinephrine metabolism, which is the core chemistry behind ADHD. It also measures 5-HIAA, the serotonin metabolite, alongside markers for mitochondrial energy production, B vitamin sufficiency, and the presence of microbial metabolites that can interfere with neurotransmitter synthesis.

The Gut-Brain Axis

The vagus nerve carries a constant stream of information between the gut and the brain. Disruption in the gut, from dysbiosis, overgrowth of certain bacteria or yeast, or increased intestinal permeability, creates systemic inflammation that reaches the brain and impairs cognitive function.

Both the Gut Zoomer and GI-MAP assess this territory using PCR-based technology, but they answer different questions. The Gut Zoomer casts a wider net across more than 300 species and excels at characterizing the overall microbial terrain, permeability, and immune markers. The GI-MAP goes deeper on

specific pathogens, offers detailed H. pylori virulence factor analysis, and includes digestive function markers like elastase and calprotectin. The clinical presentation guides which one to order. When the primary picture is microbiome dysbiosis and permeability driving neurological symptoms, Gut Zoomer is the better fit. When there is significant pathogen suspicion, active gastrointestinal symptoms, or questions about digestive enzyme output, GI-MAP gives more actionable data. There is no meaningful reason to run both at the same time.

Certain Clostridial bacteria produce metabolites that appear to interfere with neurotransmitter pathways in the brain, and yeast overgrowth produces compounds like tartaric acid and arabinose that may disrupt mitochondrial energy metabolism. These are clinically observed patterns that show up on organic acid testing and have plausible biochemical explanations, though the research establishing direct causation is still developing. That is the honest state of it, and here is what I do with it: I address those findings as part of a broader gut restoration protocol, because clinically that work frequently travels with meaningful improvements in focus and behavior, and because the interventions involved are low-risk whether or not the causal story turns out to be as clean as it looks.

Heavy Metals and Toxic Burden

Lead is a potent neurotoxin that disrupts dopamine signaling and has been found at elevated levels in children with ADHD at a disproportionate rate. Mercury, whether from dental amalgams or dietary sources, impairs methylation and interferes with neurological function in ways that can look exactly like ADHD. Aluminum, arsenic, and cadmium each have their own profiles of cognitive disruption.

Mycotoxins, the toxins produced by mold, are increasingly recognized as a driver of neurological symptoms including brain fog, poor memory, inability to concentrate, and emotional dysregulation. Many people with apparent treatment-resistant ADHD have a significant mycotoxin burden that has never been tested for.

Lab Panels I Run

I run comprehensive lab testing through Vibrant Wellness, Genova Diagnostics, and Diagnostic Solutions. A results review walks through every marker in detail and connects the findings to a specific protocol rather than a generic ADHD template. If you want to talk through which panel makes sense for your situation, that conversation is what the free Root Discovery Session is for.

Nutritional Supplementation

Bridging the Gap Between Diet and Function

Even a clean diet leaves gaps, especially in the context of ADHD, where neurotransmitter demand is higher, stress depletes nutrients faster, and gut dysfunction impairs absorption. Targeted supplementation fills those gaps and supports the biochemical pathways that diet alone cannot always reach.

Broad vitamin and mineral formulas have been tested in randomized, placebo-controlled trials in ADHD, in both children and adults, and the results are genuinely mixed: some outcomes and some raters show benefit, others do not.^{5,6,7} I read that the way I would want you to read it. Nutrient repletion is not a stimulant substitute and it does not work uniformly. It works when you were actually short of something, and that is why I test first rather than guess.

This chapter gives you two tracks: the protocol I build in practice, and the individual nutrients you can source yourself if you are working independently. Both tracks are built on the same biochemistry. Professional formulas are often built for better bioavailability and for combinations that are difficult to replicate with off-the-shelf options.

A Word on Sourcing, So You Know How This Works

I hold no physical inventory and I do not dispense supplements out of my office. I keep a screened dispensary through Fullscript, which ships directly to you, wherever you are: us.fullscript.com/welcome/drseay. Anything I name that is not carried there is available from any reputable retailer, and I will say so when that is the case.

Root Health earns a portion of dispensary purchases. That is a real financial interest and you should know about it. It does not change what I recommend, and nothing in this book requires you to buy through me.

Track 1: The Protocol I Build in Practice

This is a combination of botanical, nutritional, and select homeopathic elements, formulated for bioavailability and dosed with my supervision rather than off a shelf.

The homeopathic piece is the part of my practice most likely to draw an eye roll, so here is where it stands. The research base does not support homeopathic remedies by the standard I apply to the omega-3 trials earlier in this book, and I am not going to pretend otherwise. What I can tell you is that I use them clinically, that what I observe is clinical observation and not trial evidence, and that the cost and the risk are both low. Remedy selection is individualized, which is exactly why it belongs in a conversation rather than in a table you order from.

Support	What to look for	Dose
Acetyl-L-carnitine	Acetyl-L-carnitine, stand-alone, commonly in 500 mg capsules	1 to 2 caps with breakfast and lunch; 1 cap at 4pm
Omega-3 fatty acids	Triglyceride form (EPA plus DHA), third-party tested, for example Nordic Naturals or similar	1 to 2 g combined EPA and DHA daily with food. Check the label for concentration per capsule. Higher doses used clinically should be individualized under a provider's guidance.
Magnesium with taurine	Magnesium glycinate with taurine. Stand-alone alternative: magnesium glycinate, 200 to 400 mg elemental at night.	1 cap before breakfast; 1 cap mid-morning; 1 cap before bed
Chlorophyll and methylation support	Chlorophyllin capsules or food-grade liquid chlorophyll	1 cap twice daily
Methylated B12 with its partners (folate, B6, TMG)	Methylcobalamin, sublingual or liposomal, 2000 mcg, taken alongside folate, B6, and TMG (trimethylglycine). Homocysteine Supreme (Designs for Health) is the closest single formula to that combination.	2000 mcg daily
Immune and lymphatic support	A multi-ingredient transfer factor or immune-modulator formula supporting immune and lymphatic function, commonly liposomal. Formulations and potency vary widely across this category.	Per label direction; commonly twice daily
Colostrum, for gut-brain immune support	A bovine colostrum or transfer-factor supplement supporting gut-associated lymphatic tissue and immune messaging	Per label direction; commonly divided between morning and bedtime
Vitamin D3 with K2	D3 with K2 MK-7, liposomal spray or softgel; 2000 to 5000 IU D3 daily	5 to 10 sprays daily, or per label to reach the target IU. Test 25-OH vitamin D and monitor, especially at the higher end of that range.

Two rows above name a category rather than a product, on purpose. The immune and lymphatic formulas and the colostrum and transfer-factor products vary enormously in what is actually in the bottle, and I would rather choose the specific one with you than send you after the wrong thing on the strength of a table. Bring it up when we talk.

Track 2: Individual Nutrients, Self-Directed

If you are working without a practitioner, the following individual nutrients target the core biochemistry of ADHD. Look for the specific forms listed, because bioavailability varies significantly between forms.

Magnesium Glycinate

Magnesium is arguably the single most important mineral for ADHD support. It calms the nervous system, supports GABA production, improves sleep quality, and is required for hundreds of enzymatic reactions including those involved in neurotransmitter synthesis. Use the glycinate or threonate form specifically. Avoid oxide, which absorbs poorly. Adults: 300 to 400 mg elemental magnesium daily, taken in divided doses. A portion before bed improves sleep quality noticeably for most people.

Omega-3 Fish Oil (EPA and DHA)

Use a triglyceride-form fish oil from a reputable source that third-party tests for heavy metals and oxidation. The ethyl ester form found in most grocery store brands is poorly absorbed. Look for a combined EPA and DHA total of about 1 to 2 grams daily for general support. Nordic Naturals, Thorne, and Carlson are reliable brands. Store in the refrigerator after opening. Higher, therapeutic-range doses are used clinically for some patients; those should be individualized under a provider's guidance rather than self-started.

Zinc Picolinate or Bisglycinate

Target 15 to 30 mg elemental zinc daily, taken with food to avoid nausea. Do not take it on an empty stomach. Long-term zinc use at higher doses requires copper monitoring, since these two minerals compete for absorption. A 30 mg zinc to 2 mg copper ratio is a reasonable maintenance approach for adults.

Vitamin D3 with K2

Vitamin D deficiency is extremely common and is associated with mood symptoms, poor focus, and immune dysfunction. Test your level before supplementing if you can. For most adults with documented deficiency or insufficiency, 3000 to 5000 IU of D3 daily alongside 100 to 200 mcg of K2 MK-7 is appropriate. K2 helps keep calcium out of soft tissues and in bones where it belongs.

Methylated B Complex

Many people with ADHD carry variants in the MTHFR gene and related methylation enzymes that impair their ability to convert folic acid and B12 into the active forms. For these individuals, standard B vitamins may do very little, and some people report feeling worse on them. The active, methylated forms are the ones to look for: methylcobalamin (B12), methylfolate (5-MTHF), pyridoxal-5-phosphate (B6), and riboflavin-5-phosphate (B2). Look for a B complex that specifically uses these forms. Thorne Basic B and Pure Encapsulations B Complex Plus are solid options. The anchor dose I use is 2000 mcg of methylated B12, taken alongside folate, B6, and TMG (trimethylglycine); Homocysteine Supreme (Designs for Health) is the closest single formula to that combination.

L-Tyrosine

Tyrosine is the direct amino acid precursor to dopamine and norepinephrine. Supplementing it provides raw material the brain uses to manufacture these neurotransmitters. Take 500 to 1000 mg on an empty stomach in the morning. Some people do better with NALT (N-acetyl-L-tyrosine). It works best when other foundational deficiencies have been addressed first, particularly B vitamins and iron, which are cofactors in the conversion process.

Iron, Only If Deficient

Iron is a critical cofactor in dopamine synthesis, and low ferritin, including levels many labs still report as normal, has been associated with worse ADHD symptoms, particularly inattention. This is a case where testing before supplementing is genuinely important. Excess iron is harmful. If your ferritin is low, iron bisglycinate is the best-tolerated form. Work with a provider to monitor levels and calibrate the dose.

Phosphatidylserine

Phosphatidylserine is a phospholipid that makes up part of the outer membrane of neurons and plays a role in the transmission of nerve signals. It has been studied at 200 to 400 mg daily for cognitive performance, memory, and behavior in ADHD. It also blunts the cortisol response to stress, which is relevant to the emotional dysregulation piece. Sunflower-derived sources are preferred over soy-derived.

Probiotics

Not all probiotics are created equal, and a generic grocery store probiotic is unlikely to move the needle on gut-brain axis function. Look for multi-strain products with documented clinical research, including *Lactobacillus rhamnosus*, *Lactobacillus acidophilus*, and *Bifidobacterium longum* at a minimum. *Saccharomyces boulardii* is a beneficial yeast that specifically helps with gut permeability and microbial balance. Brands with third-party verification of CFU counts at time of expiration are worth choosing.

A Note on Supplement Quality

The supplement industry is not well regulated. Third-party testing for purity and potency matters. For individual nutrients, look for NSF Certified for Sport, USP Verified, or Informed Sport certification. Brands that consistently meet these standards include Thorne, Pure Encapsulations, Integrative Therapeutics, and Designs for Health.

Order your effort the same way you would order your attention. Build the food foundation first, then add nutrients that testing says you are actually short of. A protocol built on your own lab results will outperform a shelf of excellent supplements aimed at a problem you did not have.

Therapeutic Modalities

Tools That Retrain the Nervous System

Supplements and diet address the biochemistry underneath ADHD. The modalities in this chapter work at the level of the nervous system itself. Several of them have a decade or more of clinical research behind them, and several are part of how I work with ADHD every week.

These are not gimmicks. They are tools that, used consistently and combined with the nutritional and dietary foundation from earlier chapters, can produce meaningful and lasting changes in how the brain regulates attention, emotional tone, and executive function.

BrainTap

BrainTap uses light and sound stimulation delivered through a specialized headset to guide the brain toward specific brainwave states. The technology combines binaural beats, isochronic tones, guided visualization, and light pulses through closed-eye glasses, with the goal of shifting the brain out of the high-frequency beta states associated with stress and scattered attention and into more organized alpha and theta states associated with relaxed focus, learning, and recovery. Binaural beats and brainwave entrainment have a body of research behind them. BrainTap as a specific platform is newer to clinical study, so the evidence base is still growing.

For someone with ADHD, whose prefrontal cortex tends to be underactivated and whose brainwave patterns often show more slow-wave activity than average, tools that guide the brain toward more organized states are a rational complement to other interventions. The proposed mechanism is that repeated exposure to these guided states helps the brain access them more readily over time. Clinically, many of my patients report improved sleep, less mental restlessness, and a greater sense of calm focus with consistent use, though individual responses vary. That is clinical observation, not trial evidence, and I would rather label it than dress it up.

Sessions typically run 20 to 30 minutes and can be done at home. The library includes sessions designed for focus, sleep, stress reduction, and learning. Consistency matters more than intensity. Five to seven sessions per week over at least 90 days is where I have seen the clearest benefit.

Vagal Nerve Stimulation

The vagus nerve is the longest cranial nerve in the body and the primary communication highway between the gut and the brain. In people with ADHD, vagal tone tends to be low. Low vagal tone means a nervous system that is slow to shift from stress states back to calm, which is exactly what shows up as emotional dysregulation, poor impulse control, and difficulty transitioning between tasks.

Devices that stimulate the vagus nerve, whether through gentle electrical stimulation at the ear, through the neck, or through breathing techniques that activate the vagal response indirectly, have shown promising results for attention, emotional regulation, and anxiety. The Nurosyl and Pulsetto devices are consumer-grade auricular stimulators with growing research behind them. Apollo Neuro is a wearable that uses vibration patterns shown to increase heart rate variability, a direct measure of vagal tone. None of these are stocked or dispensed through my office; they are purchased directly from the manufacturer or from any reputable retailer.

Outside of devices, vagal activation through slow diaphragmatic breathing is free and immediately accessible. A simple practice of five-count inhalations and seven-count exhalations, practiced for ten minutes daily, measurably improves vagal tone over time and is a legitimate therapeutic tool on its own. If you take one thing from this chapter and spend nothing, take that.

Neurofeedback

Neurofeedback is one of the most researched non-pharmacological interventions for ADHD. It uses real-time EEG monitoring to show the brain its own activity patterns and reward it for producing more organized, focused output. Protocols designed for ADHD typically target the theta to beta ratio, training the brain to produce more beta activity in the prefrontal cortex.

The evidence base here is stronger than for most non-drug options. Meta-analyses report improvements in inattention, hyperactivity, and executive function, with some evidence that gains persist after the training course ends, which is not how a medication effect behaves once the medication stops. How large and how durable those gains are is still argued in the literature, particularly in the most tightly blinded trials, and I would rather you hear that from me than find it later and wonder what else I skipped. The standard treatment course is 30 to 40 sessions, usually administered twice weekly, and results build over the course of treatment.

Infrared Sauna

Infrared sauna is relevant to ADHD primarily through two mechanisms: detoxification and nervous system regulation. Far-infrared heat penetrates deeper into tissue than conventional sauna heat, mobilizing fat-soluble toxins including heavy metals, pesticides, and mycotoxins for excretion through sweat. For people whose ADHD symptoms are being driven or worsened by toxic burden, regular sauna use is a meaningful part of a detoxification protocol.

The parasympathetic activation that follows a sauna session also supports better sleep and nervous system recovery. Many people with ADHD have chronic sleep disruption that compounds every cognitive symptom, and objectively measured sleep in ADHD shows that sleep continuity differences are common, though the patterns vary considerably from person to person.⁹ Sauna use three to four times per week, particularly in the early evening, is reported to improve both how quickly people fall asleep and how well they sleep.

Infrared sauna blankets are a more accessible option for home use. They fall short of a full sauna session, though they still provide meaningful infrared exposure and are a reasonable starting point.

Red Light Therapy

Photobiomodulation, or red light therapy, uses specific wavelengths of red and near-infrared light to stimulate mitochondrial function in cells. The cytochrome c oxidase enzyme in the mitochondrial electron transport chain absorbs these wavelengths and uses them to enhance ATP production.

For brain function this matters because neurons are among the highest energy-consuming cells in the body and are acutely sensitive to mitochondrial dysfunction. Emerging research on transcranial photobiomodulation, delivering near-infrared light through the skull, has shown early promise for attention, working memory, and executive function in small studies. This is a developing area and the evidence is still preliminary. What keeps it on my list anyway is that the mechanistic rationale is sound and the safety profile is favorable, which makes it a reasonable thing to trial against your own baseline rather than a reason to overhaul a protocol. Panel devices positioned close to the head, or dedicated transcranial devices, are the common delivery methods. Daily sessions of 10 to 20 minutes are the typical protocol used in research settings.

Cold Water Exposure

Brief cold exposure, through cold showers or a cold plunge, activates the sympathetic nervous system acutely and then produces a strong parasympathetic rebound. Research has documented substantial increases in norepinephrine following cold water immersion. The direct translation of that to ADHD symptom management is extrapolated rather than proven in clinical trials, so what follows is reasoning from mechanism. For someone with chronically low norepinephrine availability, short-term elevation through cold exposure is a physiologically reasonable tool inside a broader protocol, and it costs nothing to test on yourself.

The practical application is a cold finish at the end of a warm shower, two to three minutes. Consistent practice over weeks builds tolerance and may gradually improve baseline regulation of these systems. As part of a comprehensive protocol rather than a stand-alone treatment, cold exposure is useful and accessible.

Movement and Exercise

Exercise is a hard intervention in functional medicine rather than a soft recommendation, and it has the best trial evidence of anything in this chapter. Systematic reviews and meta-analyses of structured exercise in ADHD report benefits across symptom and executive function outcomes in children and adolescents, and reviews in adults with ADHD report benefits for inhibitory control specifically.^{10,12} Aerobic exercise increases dopamine, norepinephrine, and BDNF (brain-derived neurotrophic factor) in ways that are directly relevant to ADHD. A single bout of moderate-intensity exercise improves executive function for hours afterward.¹³ Regular training produces lasting changes in the prefrontal cortex.

The most effective exercise for ADHD appears to be moderate-intensity aerobic work done consistently. Running, cycling, rowing, swimming, and jump rope all qualify. Thirty to forty-five minutes, four to five times

per week, is the target I use. Doing it in the morning, or at minimum before any period that demands high cognitive output, gets you the most out of it.

Martial arts, rock climbing, and other activities that require simultaneous physical and cognitive engagement appear to produce even stronger benefits, likely because they train executive function directly alongside the physiological adaptations. Reviews of cognitively engaging physical activity in ADHD support that pattern.¹¹

Building Your Modality Stack

You do not need all of these at once. A realistic and effective starting stack looks like this:

1. Daily 30-minute aerobic exercise, morning preferred
2. Vagal breathing practice, 10 minutes daily (five-count inhale, seven-count exhale)
3. BrainTap or neurofeedback sessions, five to seven times per week
4. Infrared sauna three to four times per week if accessible
5. A cold finish to the morning shower, daily

Stack these on top of the dietary changes and supplementation from earlier chapters. Each layer amplifies the others.

Working With a Practitioner

Why Personalization Changes Everything

Everything in this guide is true. The dietary principles, the supplements, the testing, the modalities: all of it is grounded in real physiology, and I have told you plainly where the evidence is strong and where it is thin.

What a book cannot do is know your labs. A practitioner trained in functional medicine takes this framework and applies it to your specific biochemistry, your specific history, your specific results. For people with complex or treatment-resistant presentations, that difference is often the whole difference.

What This Looks Like at Root Health

At Root Health in Garner, North Carolina, I work from a system I call the M4 Method, for Movement, Manipulation, Medicinal Nutrition, and Metabolic Connection. That is how I look at every patient, as a whole person with interconnected systems rather than a collection of symptoms to be suppressed.

For ADHD, the process typically moves through several phases.

Phase 1: Assessment and Foundation

The first step is always understanding what is happening. That means a detailed health history, a physical exam, and comprehensive lab testing. That is the assessment. Objective lab data and a careful history determine the clinical picture and the order of operations, and nothing else substitutes for them.

Foundation work begins at the same time: cleaning up the diet, supporting drainage and elimination pathways, supporting gut health, and making sure the body is not in a state where it cannot process what I give it. Trying to address ADHD-specific pathways before the foundational systems are functional is like pouring water into a cracked cup.

Phase 2: Targeted Intervention

Once the foundation is stable, I target the specific systems the testing points to. That might be neurotransmitter pathway support, reducing heavy metal burden, gut microbial correction, methylation support, or some combination. Protocols are built around your actual lab findings.

Phase 3: Restoration and Maintenance

As the imbalances correct, the work shifts toward restoring function and building resilience. This is where the modalities in the previous chapter do their heaviest lifting: neurofeedback to train new brainwave patterns, BrainTap to support cognitive reorganization, and the exercise and lifestyle practices that hold the gains made in earlier phases.

What to Expect When You Reach Out

The entry point is the Root Discovery Session, and it is free. It maps your history, identifies the key systems involved in your presentation, reviews any existing labs, and builds an initial roadmap for your care.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

What Improvement Actually Looks Like

I am not going to give you a timeline, because I would be making it up. What I will tell you is what I watch for.

Early on, when foundational nutritional gaps get corrected, what usually shifts first is the baseline, with fewer crashes, steadier afternoons, and sleep that restores something. As gut function improves and toxic burden comes down, what tends to follow is cognitive clarity and emotional regulation, the parts that are hardest to force with effort alone. Further out, for people who stay consistent, the report I hear most often is that their neurology stopped fighting them and they can work with it rather than against it, which is a different thing from ADHD disappearing.

You will notice I did not attach numbers to any of that. That is deliberate. The only measurement that means anything for you is your own log compared against your own baseline, which is why the tracker at the end of this book matters more than any average I could quote you.

This is not a quick fix. Functional medicine never promises that. What it offers is an honest look at why the symptoms are there and a systematic process for addressing the actual drivers. For people who have tried everything else, that honesty is often the most valuable part.

Putting It All Together

A 90-Day Starting Roadmap

The chapters above contain a lot of information. This final chapter distills it into a sequence you can actually implement. You do not have to do everything at once. This is a framework, not a checklist. Start where you can and build from there.

Weeks 1 to 4: The Foundation

- Remove refined sugar, artificial dyes, and processed foods from the diet immediately
- Begin building meals around protein, vegetables, healthy fats, and low-glycemic complex carbohydrates
- Start magnesium glycinate (300 mg at night to start) and a high-quality omega-3 (1 to 2 g combined EPA and DHA daily)
- Begin a methylated B complex if you have not already
- Add 30 minutes of aerobic exercise daily, or at minimum five days per week
- Begin a diaphragmatic breathing practice, 10 minutes per day
- Start the tracker at the end of this book before you change anything else, so you have a real baseline
- If possible, get your foundational labs ordered: at minimum an Organic Acids Test and a Micronutrient Panel

Weeks 5 to 8: Expanding the Protocol

- Add phosphatidylserine (200 to 400 mg daily) once the foundational supplements are established
- If testing has returned, begin targeted support based on actual findings
- Add zinc picolinate (15 to 30 mg with food) if not already included
- Begin BrainTap sessions, or look into neurofeedback in your area
- Consider adding infrared sauna two to three times per week if accessible
- If gut testing shows dysbiosis or overgrowth, begin a targeted gut protocol

Weeks 9 to 12: Deepening the Work

- Review your tracker honestly against your week one baseline, noting what has improved and what has not shifted at all
- If you are working with a practitioner, this is the phase where protocols get refined based on your actual response

- Increase BrainTap or neurofeedback frequency if you tolerate it well
- Begin addressing toxic burden if testing revealed heavy metals or mycotoxins
- Treat sleep hygiene as a specific therapeutic target, with a consistent bedtime, a room at 65 to 68 degrees, no screens in the hour before bed, and magnesium before sleep. Behavioral sleep work has the clearest research support of the sleep interventions studied in ADHD.⁸

The people who get the best results from this kind of work are the ones who start, adjust, keep going, and stay curious about what their body is telling them, whether or not they began with the most information or the most resources. The data you gather from your own experience, especially combined with objective lab testing, is genuinely powerful.

This guide is a beginning, not an endpoint. The next step is the one that moves you forward.

Final Thought

ADHD is a pattern of dysregulation with specific drivers in specific people, rather than a character flaw or a life sentence, and those drivers can be found, addressed, and in many cases meaningfully improved. The road looks different for everyone, but the destination is a version of yourself that is clearer, more regulated, and more able to access the focus and energy that are already in there.

That is what I work toward with every patient. I would be honored to be part of your story.

Your ADHD Tracker

Rate each item once daily for at least two weeks before you change anything, so you have a real baseline instead of a memory of one. Run the same tracker again at 30 and 90 days into any new approach. Gradual change is almost impossible to judge from memory, which is exactly why this page exists.

Date	Hours slept	Sleep quality (1 to 5)	Focus (1 to 10)	Restlessness (1 to 10)	Emotional regulation (1 to 10)	Protein at every meal (Y/N)	Movement today (Y/N)	Notes

Working With Root Health

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

The Root Discovery Session is the starting point for functional medicine care with me. It is a thorough look at your health history, your current presentation, and your lab findings, and it is free with no obligation. You can also reach the office directly at **(919) 662-0520**.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

I run testing through Vibrant Wellness, Genova Diagnostics, and Diagnostic Solutions. Results are reviewed in detail, connected to your specific symptoms, and used to build a protocol that addresses your actual drivers rather than a generic ADHD template.

Not ready for a conversation yet? Start tracking.

The tracker above works on paper. The digital version does the same job on your phone, keeps your history in one place, and shows you connections that are hard to see from the inside: how sleep tracks against next-day focus, and whether mood and attention move together or separately. Two weeks of honest data makes any future appointment, with me or with your own physician, far more useful than memory alone.

Start your free tracker at rootcauseunlocked.com/tracker.

Root Health | Garner, North Carolina rootcauseunlocked.com

Quick Reference

Key Lab Tests Summary

Lab test	Lab company	What it reveals	Why it matters for ADHD
Organic Acids Test	Vibrant Wellness	Neurotransmitter, mitochondrial, gut, and nutrient markers	The first place I look for ADHD biochemistry
Gut Zoomer, choose one: microbiome terrain	Vibrant Wellness	Best for microbiome ecology, gut permeability, and barrier assessment	Choose when microbiome terrain and neuroinflammation are the primary concern
GI-MAP, choose one: pathogen precision	Diagnostic Solutions Lab	Best for specific pathogens, H. pylori detail, parasites, and digestive function markers	Choose when pathogen suspicion or GI symptoms are driving the neurological picture. Run one or the other, not both.
Micronutrient Panel	Vibrant Wellness / Genova	Intracellular nutrient status across more than 40 markers	Finds your actual deficiencies
Total Tox Burden	Vibrant Wellness	Heavy metals, mycotoxins, and environmental chemicals	Toxic burden assessment
Neural Zoomer Plus	Vibrant Wellness	Neurological antibody panel, blood-brain barrier markers	Neuroinflammation assessment

Protocol Summary

Support	What to look for	Dose
Acetyl-L-carnitine	Acetyl-L-carnitine, commonly 500 mg capsules	1 to 2 caps with breakfast and lunch; 1 cap at 4pm
Omega-3	Triglyceride form (EPA plus DHA), third-party tested	1 to 2 g combined EPA and DHA daily with food; check label concentration. Higher clinical doses should be individualized under a provider's guidance.
Magnesium	Magnesium glycinate with taurine; or magnesium glycinate alone, 200 to 400 mg elemental at night	1 cap before breakfast; 1 mid-morning; 1 before bed
Methylation and chlorophyll	Liquid chlorophyll or chlorophyllin capsules	1 cap twice daily
Methyl B12 with its partners (folate, B6, TMG)	Methylcobalamin, sublingual or liposomal, 2000 mcg, alongside folate, B6, and TMG. Homocysteine Supreme (Designs for Health) is the closest single formula.	2000 mcg daily
Immune and lymphatic support	Multi-ingredient transfer factor or immune-modulator formula, commonly liposomal. Formulations and potency vary widely.	Per label direction; commonly twice daily
Vitamin D3 with K2	D3 with K2 MK-7, liposomal spray or softgel; 2000 to 5000 IU D3 daily	5 to 10 sprays daily, or per label to reach the target IU; test 25-OH vitamin D and monitor

Brands Worth Knowing for Individual Nutrients

- **Thorne Research.** A full line of methylated B vitamins, magnesium, D3 with K2, zinc, and amino acids
- **Pure Encapsulations.** Hypoallergenic formulas, strong quality control, and a wide range
- **Nordic Naturals.** Triglyceride-form fish oil, third-party tested for oxidation and metals
- **Integrative Therapeutics.** Clinical-grade formulas widely used in functional medicine
- **Designs for Health.** Practitioner-grade formulas with strong clinical backing
- **Jarrow Formulas.** A budget-conscious option with solid quality for basic nutrients

Always verify that any supplement you buy has third-party testing for purity and potency. The brands above have established track records, but the specific product you choose within any brand still deserves a label review. Most of these lines are carried on Fullscript, which ships directly to you:

us.fullscript.com/welcome/drseay. Root Health earns a portion of dispensary purchases.

This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment. It does not diagnose ADHD or any other condition. It is not a substitute for a relationship with your physician or other qualified health care provider. Never stop, start, reduce, or otherwise change the dose of any prescription medication, including stimulant medication, without first talking to the prescriber managing your care. Always consult your physician before starting any new supplement, particularly if you are pregnant, nursing, taking prescription medications, or being treated for another condition. If you are having thoughts of suicide or self-harm, call or text 988, call 911, or go to your nearest emergency department immediately.

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Further Reading

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